📘 LLM Inference Parameters – A Simple & Practical Guide

MicroDegree – Gen AI Developer

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1. What are LLM Inference Parameters? (Big Picture)

When you send a prompt to an LLM, you are not just asking a question.
You are also **controlling how the model thinks and responds**.

👉 **Inference parameters** are **knobs** that control:

- How **creative** the answer is
- How **random or predictable**
- How **long or short**
- How **focused or exploratory**

💡 **Training = learning knowledge**
💡 **Inference = deciding how to answer**

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2. Mental Model (Very Important)

Think of an LLM like a **smart storyteller**.

Inference parameters decide:

- Does the storyteller **stick strictly to facts**?
- Or **improvise creatively**?
- Does it **finish quickly**?
- Or **talk at length**?

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3. Core Inference Parameters (Must Know)

Parameter	Controls	Simple Meaning

`temperature`	Creativity	How wild the answers can be
`top_p`	Word selection range	How many possible words are allowed
`max_tokens`	Output length	How long the answer can be
`presence_penalty`	Topic repetition	Encourage new ideas
`frequency_penalty`	Word repetition	Avoid repeating same words
`stop`	Stop conditions	When to stop generation

We'll go **one by one with examples**.

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4. Temperature (Most Important)

🔹 What is Temperature?

Temperature controls **randomness**.

- **Low temperature (0–0.3)** → Safe, factual, predictable
- **Medium (0.5–0.7)** → Balanced
- **High (0.8–1.2)** → Creative, risky, imaginative

🔹 Simple Analogy

☕ Cold coffee → Stable
🔥 Hot coffee → Wild

🧪 Example Prompt

```
Explain what AI is.
```

Temperature = 0.1 (Very Low)

```
Artificial Intelligence is the simulation of human intelligence in
machines that are programmed to think and learn.
```

✅ Accurate
❌ Boring
❌ No creativity

Temperature = 0.7 (Balanced)

```
Artificial Intelligence refers to machines designed to mimic human
thinking, learning from data to solve problems like humans do.
```

✅ Clear
✅ Slightly expressive
✅ Best for production

Temperature = 1.0 (High)

```
Artificial Intelligence is like giving machines a brain, allowing them
to learn, adapt, and sometimes surprise us with how human-like they
behave.
```

✅ Creative
❌ May exaggerate
❌ Less precise

🎯 When to Use

Use Case	Temperature
Legal, Medical, Compliance	0.1–0.3
Chatbots, Education	0.5–0.7
Marketing, Storytelling	0.8–1.0

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5. Top-P (Nucleus Sampling)

🔹 What is Top-P?

Instead of choosing from **all words**, the model chooses from the **most probable words** whose total probability = `top_p`.

- `top_p = 1.0` → All words allowed
- `top_p = 0.9` → Only top 90% probable words

🔹 Simple Analogy

🍽️ Buffet:

- Top-P = 1 → Eat everything
- Top-P = 0.8 → Only best dishes

🧪 Example Prompt

```
Write a motivational quote.
```

top_p = 0.3

```
Success comes from consistent effort and discipline.
```

✅ Safe
❌ Generic

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top_p = 0.9

```
Success grows when you show up daily, even on days when motivation
feels absent.
```

✅ Rich
✅ Natural language

🎯 Best Practice

- Use **either** `temperature` **or** `top_p`
- Don't aggressively tune both together

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6. Max Tokens (Output Length Control)

🔹 What are Tokens?

- Tokens ≠ words
- Roughly: **1 token ≈ 0.75 words**

🔹 What max_tokens Does

Controls **how long the response can be**.

🧪 Example Prompt

```
Explain Machine Learning.
```

max_tokens = 20

```
Machine Learning enables systems to learn from data and improve
without explicit programming.
```

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max_tokens = 150

```
Machine Learning is a subset of Artificial Intelligence that allows
systems to learn from data, identify patterns, and make decisions with
minimal human intervention. It improves performance over time through
experience.
```

🎯 Best Practice

Use Case	max_tokens
Short answers	50–100
Explanations	200–400

Blogs	800–1500

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7. Presence Penalty (New Topics)

🔹 What it Does

Encourages the model to **introduce new ideas** instead of staying on the same topic.

- Range: `-2.0` to `2.0`
- Higher → More topic diversity

🧪 Example Prompt

```
List benefits of AI.
```

presence_penalty = 0

```
AI improves efficiency, automation, and accuracy.
```

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presence_penalty = 1.0

```
AI improves efficiency, enables personalization, accelerates research,
enhances decision-making, and opens new business models.
```

🎯 Use When

- Brainstorming
- Idea generation
- Content expansion

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8. Frequency Penalty (Repeat Words Control)

🔹 What it Does

Penalizes **repeated words**.

🧪 Example Prompt

```
Describe AI in simple terms.
```

frequency_penalty = 0

```
AI is a system that learns from data. AI helps machines think. AI
improves automation.
```

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frequency_penalty = 1.0

```
AI refers to systems that learn from data, helping machines perform
tasks intelligently and improve automation.
```

🎯 Use When

- Avoid repetitive chatbot responses
- Improve writing quality

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9. Stop Sequences

🔹 What it Does

Stops generation when a specific token appears.

🧪 Example

```
stop = ["###"]
```

Output stops when `###` appears.

Used in:

- Structured outputs
- Multi-step prompts
- Tool calling

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10. Real-World Parameter Presets (Very Important)

🔹 Chatbot (Production Safe)

```
{
  "temperature": 0.6,
  "top_p": 0.9,
  "max_tokens": 300,
  "frequency_penalty": 0.3,
  "presence_penalty": 0.2
}
```

🔹 RAG / Knowledge System

```
{
  "temperature": 0.2,
  "top_p": 1.0,
  "max_tokens": 400
}
```

🔹 Creative Writing

```
{
  "temperature": 0.9,
  "top_p": 0.95,
  "max_tokens": 800
}
```

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11. Common Beginner Mistakes 🚨

❌ Using `temperature = 1.5` for factual answers
❌ Setting `max_tokens` too low → incomplete answers
❌ Tuning all parameters at once
❌ Expecting same output every time with high temperature

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12. Key Takeaways (Exam Ready)

- **Temperature = creativity**
- **Top-P = vocabulary scope**
- **Max Tokens = length**
- **Penalties = repetition control**
- **Inference tuning is as important as prompting**

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13. One-Line Summary (Interview Ready)

> *Inference parameters control how an LLM responds, not what it knows. Proper tuning balances accuracy, creativity, and reliability based on the use case.*